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RESEARCH ARTICLE

Semi-Supervised Learning With Wafer-Specific Augmentations for Wafer Defect Classification

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ABSTRACT Semi-supervised learning (SSL) models, which leverage both labeled and unlabeled datasets, have been increasingly applied to classify wafer bin map patterns in semiconductor manufacturing. These models typically outperform supervised learning models in scenarios where labeled data are scarce. However, the challenges posed by wafer bin map data in applying conventional image augmentation methods, particularly within SSL frameworks such as FixMatch, have not yet been fully addressed despite their significant role. Recognizing the importance of thoughtful implementation of weak and strong augmentations within FixMatch, we propose a method that incorporates saliency map information into cutout augmentation. This approach preserves essential regions crucial for wafer defect pattern classification and thus improving model performance. Our approach achieves a macro F1-score of 0.841 with only 5% labeled data, surpassing state-of-the-art methods by 6.2% compared to WaPIRL and 7.5% compared to Manivannan's method. Similarly, with 10%, 25%, and 50% labeled data, our method achieves F1-scores of 0.856, 0.874, and 0.891, respectively, showing improvements of 3.5%, 1.7%, and 1.2% over WaPIRL and 5.0%, 6.6%, and 11.9% over Manivannan's method in each case. Experimental results indicate significant improvements in defect pattern classification by avoiding cutting important regions in cutout augmentation. The proposed method achieves new state-of-the-art performance in wafer bin map defect classification, demonstrating the potential of our tailored augmentation techniques and the effectiveness of incorporating saliency map information reflecting the characteristics of wafer bin maps.

INDEX TERMS Data augmentation, defect patterns classification, semiconductor manufacturing, semi-supervised learning, wafer bin maps.

I. INTRODUCTION

In semiconductor device manufacturing, a wafer is a thin disk made of silicon that comprises hundreds of semiconductor chips. It undergoes a series of intricate processes designed to create integrated chips on the silicon wafer. Before packaging, every wafer undergoes multiple rounds of electrical tests to determine whether each die (or chip) meets the product specifications. The wafer bin map (WBM) is a two-dimensional binary map that reflects the results of these electrical tests in which failed dies are assigned a value of one and passed dies are assigned a value of zero. From this map, the location and shape of the failed dies are noted to perform WBM defect pattern classification, a crucial process for improving the overall yield in semiconductor manufacturing.

Typically, these defect patterns are categorized into two types: global random defects and local systematic defects. Global random defects appear to randomly scatter across the entire WBM, even under normal production conditions [1]. In contrast, local systematic defects exhibit spatial correlations in specific areas of the wafer, creating patterns such as central circles, edge rings, localized areas, and scratches [2].

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These local defects are typically associated with specific process issues and can be easily resolved by experienced process engineers once identified, thereby identifying the root cause of the failures. Defects that can arise from various factors, such as impurities, equipment failures, and process variations, can adversely impact the functionality and reliability of the device. Therefore, improved automated methods capable of accurately identifying these local defect patterns are required [2].

Obtaining a large amount of labeled data for classification models is not only time-consuming and costly, but also requires expert knowledge. Accurately labeling data requires the discerning eye of a domain expert, which further increases the time, cost, and complexity of the process. In addition, the availability of experts can frequently be a limiting factor, creating bottlenecks in the data labeling process. This can significantly slow down the training of machine learning models, delaying the delivery of valuable insights and solutions.

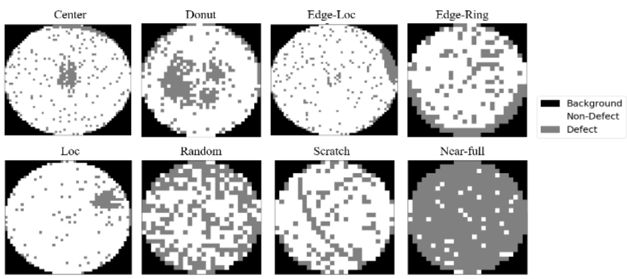

FIGURE 1. Schematic overview of WBM samples with eight defect patterns.

To overcome this challenge, self-supervised and semi-supervised learning (SSL), leveraging a wealth of unlabeled data, have emerged as potential solutions. These methods have been effectively applied across diverse fields, including image and video classification, fault detection, medical diagnosis, and natural language processing [3], [4], [5], [6], [7]. Despite their widespread applications, only a few studies have applied SSL [8], [9] or self-supervised learning [10] to WBM image classification. Previously, the primary focus was on applying new learning methods to WBM datasets using unlabeled data. However, these studies have typically overlooked the critical role of data augmentation. This is where the main challenges with these methods lie: primarily, the effective implementation of augmentation techniques.

The application of augmentation techniques suited for general images, as commonly seen in existing research, is limited to WBM datasets. Unlike typical images, WBMs are in an image data format that not only indicates the position of each chip, but also determines whether it is defective. These defect parts, which are located in small, precise areas, are crucial for defect pattern classification. Thus, information about each position is crucial. Furthermore, WBM images comprise a single channel of three-digit data, the visual representation of which is shown in Figure 1. This inherent uniqueness of WBM images necessitates the development of specialized augmentation techniques.

Applying data augmentation into SSL needs to consider consistency regularization carefully to improve model generalization in scenarios with abundant unlabeled data; however, it has not yet been customized to fit the unique characteristics of WBMs. To bridge this gap, we propose an approach for WBM data augmentation in the context of SSL. The experiments and configurations build upon the pioneering work of Kahng and Kim, who proposed a simple yet effective self-supervised learning approach for defect classification [10]. The main contributions of this study can be summarized as follows:

1) The proposed method introduce customized weak and strong augmentation techniques tailored specifically for WBM images. Weak augmentations utilize only rotation to exploit the rotation-invariant nature of WBMs, moving away from conventional horizontal and vertical flipping used in FixMatch. Strong augmentations integrate multiple affine transformations while excluding color-based augmentations, considering that WBM images are grayscale and do not benefit from RGB-specific transformations.
2) We incorporate saliency maps generated from a pretrained model to preserve critical regions of WBM images essential for defect classification. This approach ensures that the most informative areas are retained, enabling the model to focus on regions critical for identifying defect patterns effectively.

The remainder of this paper is organized as follows. Section II provides a brief review of research on WBM defect pattern classification. Section III explains the details of the proposed methodology. Section IV presents the experimental results. Finally, Section V includes the conclusions and future research directions.

II. RELATED WORKS

A. WAFER BIN MAP CLASSIFICATION

Semiconductor WBM defect pattern classification has been advanced using statistical and traditional machine learning techniques based on features, which are extracted from the knowledge of domain experts. Wang et al. [6] proposed a mixed clustering method using spatial filters to identify defect patterns. Li and Huang [11] trained self-organizing maps for defect pattern discovery using assigned clusters for support vector machine training. Kim et al. [12] developed normalized singular value decomposition to distinguish between single- and non-single-bit failure maps. Saqlain et al. [13] proposed the use of a soft voting ensemble classifier that combines density, radon, and geometric features. However, the success of this method depended on the informativeness of the handcrafted features and was confronted with the potential loss of important unique information from the raw WBM during the feature extraction process.

TABLE 1. Comparative analysis of WaFixMatch, Manivannan's and WaPIRL.

	WaFixMatch	Manivannan's [9]	WaPIRL [10]
Main Contribution	Semi-supervised learning with wafer-specific augmentations using saliency maps to preserve crucial regions.	Ensemble-based SSL using multiple CNNs with label smoothing to reduce over-confidence in predictions	Self-supervised representation learning using contrastive learning to pre-train a CNN encoder.
Training Strategy	Single-stage approach, directly using pseudo-labels and consistency regularization	Uses ensemble of CNNs and pseudo-labeling to train on labeled and unlabeled data	Two-stage approach with self-supervised pre-training on unlabeled data, followed by fine-tuning on labeled data
Augmentations	Weak and strong augmentations with saliency map guidance.	Random horizontal and vertical flipping	One or two augmentations among 5 augmentations for each scenario
Strengths	Enhances augmentations while preserving critical image areas.	Increases robustness and reliability using ensemble and label smoothing techniques	Self-supervised pre-training on unlabeled data
Weakness	Dependency on saliency maps	High resource demand due to ensemble training	Requires researchers to choose one or two optimal augmentations

With the advent of deep learning, the paradigm has shifted to allow the direct extraction of complex patterns from raw data, resulting in significant improvements in classification performance. Wang and Chen [14] used a multilayer perceptron on features extracted by a set of spatial filters manually designed to reflect the rotational invariance of WBMs. Convolutional neural networks (CNNs) were used to classify defect patterns based on mixed-type defect patterns and a synthetic WBM dataset [1]. However, these methods have limited applicability in scenarios with a shortage of labeled samples, such as during process development or initial production stages. Unsupervised learning, which does not require labels, was considered a promising strategy for developing fully automated classification tools. Conventional clustering methods, such as K-means and hierarchical clustering, have been used; recently, unsupervised deep learning algorithms have been used for pattern detection. Lee et al. [15] used a stacked denoising autoencoder for WBM feature extraction and classification. Nakazawa and Kulkarni [16] used a deep convolutional encoder–decoder neural network to detect unseen patterns. However, the performance of unsupervised learning in the absence of label guidance was frequently found to be unsatisfactory.

Because of the manual labor and time required for the labeling process, researchers began to explore self-supervised and SSL techniques. Early self-supervised learning showed promising prospects by enabling the extraction of useful features without explicit labeling. Furthermore, the potential of SSL, which can use both labeled and unlabeled data, provided an attractive strategy and offered the best of both worlds. Pioneering studies such as those by Kong and Ni [17] and Kang et al. [10] have applied SSL and self-supervised learning, respectively to WBM defect pattern classification. Specifically, Kong and Ni applied a ladder network to incorporate unlabeled data in a semi-supervised manner; however, the dataset size could have limited the scalability of the model [15]. In particular, self-supervised and SSL plays a crucial role in improving robustness and performance through augmentation techniques. However, the unique characteristics of the WBM data require a specialized approach for augmentation. Despite the advancements in SSL models, there is a lack of application to wafers in reality.

B. AUGMENTATION TECHNIQUES FOR WAFER BIN MAP DATA

Data augmentation, an extensively used regularization technique, has improved the performance of CNNs in image classification tasks. To harness the full potential of this technique, it is vital to identify label-preserving transformations that fit the domain knowledge of a given dataset. Despite its importance, prior research on appropriate augmentations for WM811K, a dataset specific to the semiconductor industry, remains limited. Yoo and Kang [18] proposed a class-adaptive data augmentation method that used class-specific augmentation techniques. Wang and Chen [14] extracted features with a set of hand-designed spatial filters reflecting the rotation-invariant properties of WBMs. Similarly, Kang [19] demonstrated that rotation-based data augmentation could improve WBM pattern classification, particularly when training data are scarce. Furthermore, Jeong et al. [20] proposed a novel rotation and flip invariant method based on the labeling rule that the WBM defect pattern are unaffected the rotation and flip of labels, achieving class discriminant performance in scarce data scenarios.

Data augmentation techniques are critical for improving the robustness and performance of SSL. Before FixMatch, many semi-supervised learning methods focused on techniques such as ensemble multiple models, pseudo-labeling

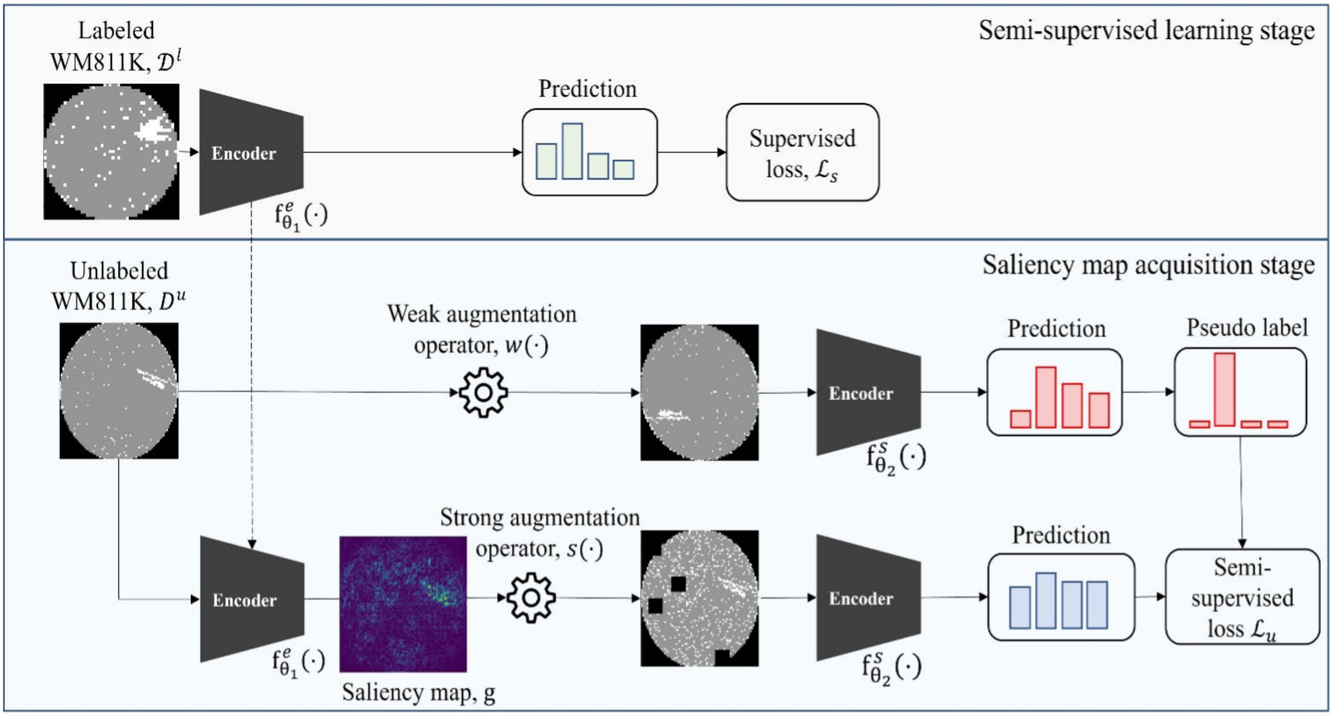

FIGURE 2. Overall structure of the proposed method, WaFixMatch.

and labelsmoothing [3]. Manivannan [9] explored approaches to mitigate over-confidence in predictions and improve stability using simple augmentations like random horizontal and vertical flipping. FixMatch, which was built on the foundation of these earlier methods, introduced a more refined approach to SSL [3]. FixMatch, one of the widely used SSL models and our foundational model, leverages the abundance of unlabeled data paired with a smaller labeled dataset for efficient deep learning model training. This model stands out because of its consistency regularization, which ensures model consistency across varied image augmentations [3]. In its workflow, FixMatch crafts a weak and strong augmentation of an unlabeled image. The pseudo-label of the weak augmentation then guides the learning of the model on the strongly augmented counterpart. Remarkably, this method has achieved exceptional outcomes across numerous datasets, including CIFAR-10/100 [21], SVHN [22], STL-10 [23], and ImageNet [24], particularly when provided with limited labeled data. The combination of simplicity and efficacy of FixMatch has firmly established its position within the realm of SSL. However, in [10], the performance of the cutout technique was found to be less effective compared to other augmentation methods. Our hypothesis posits that cutout, when used in both FixMatch and WaPIRL [10], may eliminate crucial regions pivotal for defect pattern recognition. To overcome this challenge, we integrate KeepAugment [25] for robust augmentation. This technique leverages the saliency map to highlight crucial image areas, ensuring their preservation during the augmentation process. Such a preservation-focused approach facilitates the creation of more accurate training samples.

To address a discernible gap in the literature, we introduce an SSL method specifically tailored to wafers, incorporating wafer-specific augmentations to preserve the unique characteristics of WBMs. Recognizing the influx of advanced models in SSL without any tailored application for wafers, our study serves wafer-specific augmentation techniques, drawing inspiration from the strength of SSL. A comparison of these approaches is presented in Table 1, with a detailed discussion provided in Section IV-C.

III. PROPOSED METHODOLOGY—WaFixMatch

Figure 2 provides an overview of the proposed methodology, the wafer-specific FixMatch (WaFixMatch), which primarily comprises two stages: (1) the acquisition of the saliency map from a pretrained model trained with a labeled dataset, and (2) the semi-supervised learning stage, which incorporates FixMatch with weak and strong wafer-specific augmentations. Transitioning into the semi-supervised learning stage, we use the pretrained model to extract the saliency map, which is then incorporated into the cutout augmentation process. The proposed augmentation technique preserves crucial areas in the image, enhancing classification performance.

As can be seen Figure 2, our framework comprises seven components: (1) a weak augmentation operator w(·), (2) a strong augmentation operator s(·), (3) a supervised network to obtain a saliency map $f^s_{\theta_1}(\cdot)$, (4) an encoder network $f^e_{\theta_2}(\cdot)$, (5) a supervised loss $\mathcal{L}_s$, (6) a semi-supervised loss $\mathcal{L}_u$, and

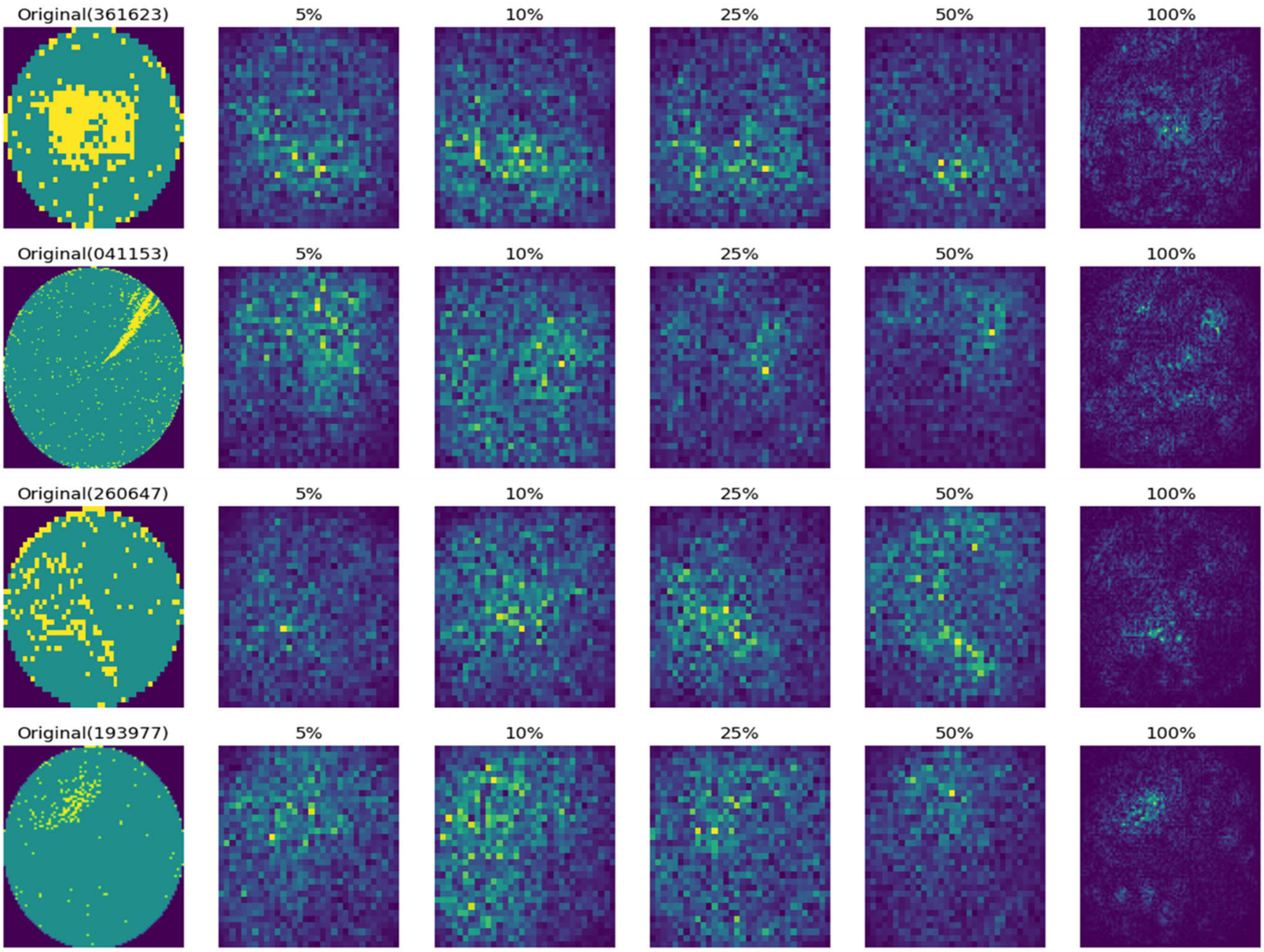

FIGURE 3. Saliency map examples for each proportion of labeled train dataset.

(7) a saliency map of a WBM image g on pixel (i, j) with the given label y, g_{ij}. Note that the subscript in $f^{e}_{\theta_2}(\cdot)$ indicates that the networks are trainable with their parameters θ_2. Denote a training dataset consisting of a labeled dataset $\mathcal{D}^l$ and an unlabeled dataset D^u. The encoder network $f^{e}_{\theta_1}(\cdot)$ is initially trained using a labeled dataset and is subsequently used to generate a saliency map for an unlabeled dataset. The generated saliency map allows us to identify areas with high saliency values, enabling us to avoid applying cutout augmentation to critical parts of the image that contain essential information about the defect or the object of interest. The weak augmentation operator w(·) is the rotation operator, and the strong augmentation operator s(·) has two augmentations, which randomly picks two among the following five choices: cropping, cutout, noise addition, rotation, and shifting [1]. Denote a training labeled dataset of $\mathcal{N}^l$ labeled WBMs as $\mathcal{W} = \{w_i\}_{i=1}^{N}$ consisting of WBMs defined as $w \in \mathbb{R}^{H \times W}$, where H and W represent the height and width, respectively.

A. SALIENCY MAP ACQUISITION STAGE

The saliency map acquisition stage aims to acquire a saliency map for subsequent augmentation of an unlabeled dataset during the semi-supervised learning stage. The saliency map measures the importance of the rectangular regions within an image and ensures that cutout augmentation avoids cutting out the regions with the highest scores. Before advancing to the semi-supervised learning stage, a CNN encoder, $f^{s}_{\theta_2}(\cdot)$, is trained with only the labeled dataset $\mathcal{D}^l$ using the following loss function:

$$\mathcal{L}_{\text{saliency}} = \frac{1}{B}\sum_{b}^{B} H\left(y_b, f^{e}_{\theta_1}(y|w(x_b))\right), \tag{1}$$

where $(x_b, y_b) \in \mathcal{D}^l$, B represents a batch size, and H is the standard cross-entropy loss on labeled samples.

Subsequently, the weights of the encoder are fixed, and the unlabeled dataset is passed through the backbone network to obtain the saliency map. This saliency map is used as a reference during the next stage, particularly when using cutout augmentation as a strong augmentation. We use a standard saliency map based on a gradient. Specifically, given an image x and its corresponding label logit value $l_y(x)$, we take $g_{ij}(x, y)$ to be the absolute value of gradients $\left|\nabla_x l_y(x)\right|$. Saliency maps for the unlabeled dataset are generated for use in the next stage, as shown in Figure 3. Gradient-based saliency maps, which calculate the gradient of the class score with respect to input pixels, highlighting the most relevant regions for a model's prediction. This method assigns higher values to important areas in the image based on their gradient values [26]. The first column in Figure 3 displays the original image, and the subsequent columns represent saliency maps obtained from an encoder that was trained with varying proportions of labeled images, specifically 5%, 10%, 25%, 50%, and 100%. These saliency maps highlight the significant regions for classifying defect patterns at varying proportion of labeled data. Unlike RGB images, WBM images do not require the channel-wise maximum approach to obtain a saliency map value for each pixel (i, j).

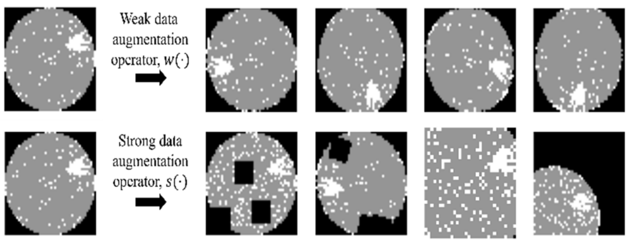

FIGURE 4. Weak and strong augmentation examples.

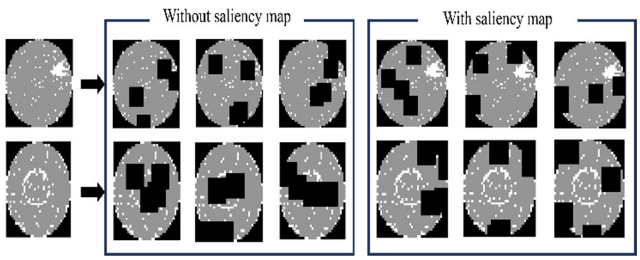

FIGURE 5. Comparison of cutout effects without and with using saliency map.

B. SEMI-SUPERVISED LEARNING STAGE

In the semi-supervised learning stage, our primary focus is to leverage the unlabeled dataset while training with the labeled dataset. The loss function for the proposed WaFixMatch consists of a supervised loss, $\mathcal{L}_s$ and an unsupervised loss, $\mathcal{L}_u$. The supervised loss, $\mathcal{L}_s$ (i.e., standard cross-entropy loss) applied to weakly augmented labeled samples can be quantified as follows:

$$\mathcal{L}_s = \frac{1}{\mathrm{B}} \sum_{\mathrm{b}}^{\mathrm{B}} \mathrm{H}\left(\mathrm{y}_{\mathrm{b,c}}, \mathrm{f}^{\mathrm{s}}_{\theta_2}(\mathrm{y}|\mathrm{w}\,(\mathrm{x}_{\mathrm{b}}))\right), \tag{2}$$

where $(x_b, \mathrm{y_b}) \in \mathcal{D}^l$ with B and H representing the batch size and the standard cross-entropy loss applied to the labeled samples. To incorporate the unlabeled images into the learning process, we initially compute the predicted class distribution for the model provided with a weakly augmented instance of a given unlabeled image. A pseudo label for D^u is thus generated by using the following equation:

$$p_b = \mathrm{f}^{\mathrm{s}}_{\theta_2}(\mathrm{y}|\mathrm{w}\,(\mathrm{u_b})), \tag{3}$$

where $\mathrm{w}(\cdot)$ represents a weak augmentation operator, and $\mathrm{u_b} \in D^u$. Subsequently, we calculate $\hat{p}_b = argmax(p_b)$ that generates a pseudo label in the form of a one-hot vector.

$$\mathcal{L}_u = \frac{1}{\mu B} \sum_{b=1}^{\mu B} I(\max\,(p_b \geq \tau_s))H\left(\hat{p}_b, \mathrm{f}^{\mathrm{s}}_{\theta_2}(y|s\,(u_b))\right) \tag{4}$$

where τ_s denotes a scalar hyperparameter indicating the threshold for retaining a pseudo-label. I is a indicator function, and μ is a hyperparameter that determines the size of D^u, which is a multiple of D^u. Consequently, the final loss function of the proposed WaFixMatch is $\mathcal{L}_s + \lambda_u \mathcal{L}_u$, where λ_u as a fixed scalar hyperparameter that determines the relative importance of the unsupervised loss.

Weak augmentation uses rotation augmentation that randomly flips images horizontally with a probability of 50% and randomly translating images by up to 12.5% vertically and horizontally. The rotation augmentation facilitates the generation of more image variations, while preserving the semantic information embedded within WBMs, particularly their circular and central architecture illustrated in the upper part of Figure 4.

For strong augmentation, the typical augmentation techniques such as RandAugment [27], CTAugment [28] might not be directly applicable to WBMs because of the unique characteristics and requirements of wafer fabrication datasets. To address this, the strong augmentation randomly chooses two sequential augmentations among cropping, cutout, noise addition, rotation, and shifting as mentioned in Section III of [10]. The lower portion of Figure 4 presents a visual representation of the resulting strong augmentation. They include cutout and noise addition, rotation and cutout, crop and noise addition, shift and noise addition sequentially.

The representation of WBMs typically consists of three digits, and their precise location at the center of the chip is critical for accurate classification and defect detection. Therefore, preserving this spatial information during data augmentation becomes crucial to maintain the essential clues required for effective defect pattern classification.

Previous data augmentation strategies such as cutout and shift may compromise critical spatial features in WBMs. Figure 5 illustrates that effects of cutout augmentation without and with using a saliency map. It is evident that cutout without using a saliency map could mask key details that deteriorate classification accuracy.

Our framework uses a robust augmentation process that incorporates a saliency map to guide cutout augmentation, ensuring the preservation of important features in WBMs. Given a region for potential removal, the saliency map helps to calculate an importance score. If the score is below a set threshold τ_c, indicating that the region is not critical, cutout proceeds; otherwise, a new region is selected. Algorithm 1 summarizes how our framework operates to preserve the characteristics of WBMs. Initially, two augmentations are randomly selected from flip, cutout, noise addition, and rotation with a priority given to cutout using saliency maps to maintain important areas. Finally, a secondary augmentation is applied for additional variation.

Algorithm 1 Pseudocode for Strong Augmentation With Saliency-Driven Cutout Augmentation

Denote original image size as (x_{orig}, y_{orig})
Denote region size for cutout as (x_{cutout}, y_{cutout})
Denote threshold for accepting cutout score as τ_c
Denote saliency map as $g(i, j)$
Initialize augmentation set $AUG = \{flip, cutout, shift, noise, rotation\}$
while Preparing Data do
  Select two augmentations (aug_1, aug_2) randomly from AUG
  if aug_1 or aug_2 is *cutout* then
   Apply cutout first:
   repeat
    Randomly choose the top-left corner (cx, cy) for the cutout region such that $cx \in [0, x_{orig} - x_{cutout}]$ *and* $cy \in [0, y_{orig} - y_{cutout}]$
    Calculate importance score for the region as: $score = \frac{1}{x_{cutout} \cdot y_{cutout}} * \sum_{i=cx}^{cx+x_{cutout}} \sum_{j=cy}^{cy+y_{cutout}} g(i, j)$
  until score $<= \tau_c$
  Execute cutout by removing the region $(cx, cy, cx + x_{cutout}, cy + y_{cutout})$
 end
 Apply the second augmentation aug_2 if aug_1 is not *cutout*; otherwise, apply aug_1
end

IV. EXPERIMENTS AND RESULTS

A. HARDWARE AND SOFTWARE CONFIGURATION

To ensure the reproducibility of our findings, we provide a detailed description of the hardware and software configurations used for our experiments. All experiments were conducted on a server equipped with NVIDIA RTX 4090Ti, 64GB RAM, Intel AMD Ryzen Threadripper PRO 3995WX 64-Cores. The code was implemented using Python 3.8 and Pytorch 1.11, and training was conducted on an Ubuntu 22.04 environment.

B. DATASETS

We evaluated the performance of the proposed WaFixMatch using the publicly available WM811K dataset (https://www.kaggle.com/datasets/qingyi/wm811k-wafer-map) [5]. The dataset consists of 811,457 samples obtained from a real-world semiconductor fabrication process. Among these samples, 172,950 were labeled, and the remaining 638,507 samples were unlabeled. The labeled samples were divided into nine predefined classes (center, donut, edge-loc, edge-ring, loc, random, scratch, near-full, none). It is noted that there was a significant class imbalance, with the "None" class representing approximately 85% of the total population. Table 2 shows detailed sample counts and proportions.

TABLE 2. Class label distribution of 172,950 labeled samples in WM811k.

Class label	Sample counts	Proportion (%)
Center	4,294	2.48
Donut	555	0.32
Edge-Loc	5,189	3.00
Edge-Ring	9,680	5.59
Loc	3,593	2.07
Random	886	0.51
Scratch	1,193	0.68
Near-Full	149	0.09
None	147,431	85.24

Because of the severe class imbalance, conventional supervised learning models were susceptible to overfitting the majority class. To address this issue, we used batch-level resampling techniques as implemented in WaPIRL [10]. Additionally, to standardize input sizes, all WBM images were resized to 96 × 96 using nearest neighbor interpolation, following the approach in WaPIRL [10]. This allowed all models to be trained on the WM-811K dataset with consistent training and validation splits.

C. BASELINES

In our experiments, we compare the proposed method with several state-of-the-art approaches in self-supervised and semi-supervised learning for wafer bin map (WBM) classification. The first baseline, WaPIRL [10], uses a self-supervised learning strategy using a contrastive loss to pre-train a CNN encoder, followed by fine-tuning with labeled data. The second baseline, Manivannan [9] uses multiple CNN models with label smoothing to address over-confident predictions, using random horizontal and vertical flips as augmentations for unlabeled data. Our method builds on these baselines by incorporating wafer-specific augmentations and leveraging saliency maps to preserve critical regions during strong augmentations. A comparison of these approaches is presented in Table 1.

WaPIRL [10] conducted experiments with four well-known CNN architectures: AlexNet, VGG16, ResNet-18, and ResNet-50. Similarly, Manivannan [9] investigated ResNet-18 and DenseNet. For a fair comparison, we initially chose the ResNet-18 architecture, which was used as an encoder in both studies. Furthermore, we extended our experiments to include ResNet-50 for WaPIRL [10] and DenseNet for Manivannan [9]. Manivannan [9] used a pretrained model from ImageNet. However, ImageNet comprised RGB images that were different from WBM images. Thus, we found it necessary to reimplement the model in the same manner as WaPIRL [10] to ensure consistency.

D. EVALUATION METRIC

We selected the F1 score as the primary evaluation metric for this study. Given the highly imbalanced nature of the dataset, where the 'None' class represents 85.24% of samples, Accuracy may not fully reflect the model's performance in identifying defect patterns. The F1 score, which balances Precision and Recall, offers a more comprehensive measure by highlighting the model's effectiveness in detecting minority classes, which is essential for our task. Additionally, previous studies in WBM classification have also utilized the F1 score to provide a more nuanced assessment, facilitating direct comparisons with existing baselines:

$$\text{F1 score} = 2\times\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}. \tag{5}$$

E. IMPLEMENTATION DETAILS

To optimize the model, we used the AdamW (adaptive moment optimizer with weight decay) optimizer [6] and the following hyperparameters: a learning rate of 0.01 and a weight decay of 0.0003. AdamW tended to be less sensitive to learning rate schedules, making it easier to tune hyperparameters and achieve optimal performance. These advantages collectively contributed to more stable and efficient training, ultimately resulting in improved performance and accuracy in wafer image classification tasks on the WBM dataset. For ResNet-18, the batch size for the input data was 256. For DenseNet-121, we kept all parameters the same but adjusted the batch size to 64, and for ResNet-50, the batch size was set to 128.

Hyperparameters tuning can be optimized more efficient by using Hyperband [7] within 70 iterations. To optimize the performance of the proposed model, we set up several hyperparameters within specific ranges based on preliminary experiments. The scaling hyperparameter, τ_s, operates within a narrow band of 0.3 to 0.5, ensuring that the model retains sensitivity within acceptable limits. τ_c, which governs the clipping threshold, is set between 0.8 and 0.95, maintaining a balance that discourages outlier influence. Similarly, the update frequency of the learning rate is controlled by λ_u, with an allowable range extending from 0.1 to a dynamic upper limit of 3.0, allowing for adaptive adjustments in response to model feedback. The integer-valued hyperparameter μ, responsible for regimen diversity in our training schedule, varies uniformly between 3 and 7, thereby injecting sufficient variability to prevent overfitting. After a series of iterative tuning sessions, the model's architecture was solidified by fixing λ at a value of 0.217, aligning with the system's natural frequency. By anchoring μ at 4, the model gains the flexibility required for nuanced pattern recognition. The τ_s parameter was precisely adjusted to 0.893 to fine-tune the model's responsiveness, while τ_c was set to 0.493 to ensure robust boundary conditions.

We ran 100 epochs in the saliency map acquisition stage and 180 epochs in the semi-supervised learning stage. We conducted experiments on five random seeds and evaluated average performance and standard deviation. Unless otherwise noted, every implementation detail equals those mentioned in [1].

TABLE 3. The classification performance reported using the macro F1 score and standard deviation (in parentheses) on WBM-811K under five different experiments with ResNet-18. The total count of labeled images used at each percentage, provided in square brackets. The best results are in bold.

Method	Proportion of Labeled Data (Number of Labeled Data Used)				
	5% [6,918]	10% [13,836]	25% [34,590]	50% [69,180]	100% [138,630]
Supervised	0.659	0.681	0.787	0.856	0.877
WaPIRL [10]	0.792	0.827	0.857	0.879	**0.895**
Manivannan's [9]	0.766	0.815	0.808	0.771	0.720
WaFixMatch (Proposed)	**0.841 (0.004)**	**0.856 (0.01)**	**0.874 (0.02)**	**0.891 (0.002)**	0.890 (0.003)

TABLE 4. The ablation study using the macro F1 score and standard deviation (in parentheses) on WBM-811K under five different experiments with ResNet-18. The total count of labeled images used at each percentage, provided in square brackets. The best results are in bold.

Method	Proportion of Labeled Data (Number of Labeled Data Used)				
	5% [6,918]	10% [13,836]	25% [34,590]	50% [69,180]	100% [138,630]
WaFixMatch (w/o KeepCutout, w/o Rotation)	0.765 (0.007)	0.813 (0.014)	0.864 (0.005)	0.879 (0.006)	0.887 (0.007)
WaFixMatch (w/o Rotation)	0.769 (0.004)	0.821 (0.013)	0.865 (0.007)	0.883 (0.007)	0.886 (0.007)
WaFixMatch (w/o KeepCutout)	0.805 (0.007)	0.839 (0.009)	0.867 (0.008)	0.879 (0.004)	0.886 (0.002)
WaFixMatch	**0.841 (0.004)**	**0.856 (0.01)**	**0.874 (0.02)**	**0.891 (0.002)**	**0.890 (0.003)**

F. TRAINING STRATEGY

Manivannan [9] uses a two-stage training process, initially training individual CNNs with labeled data and subsequently incorporating unlabeled data with pseudo-labels and weights for semi-supervised learning. Kahng and Kim adopt a self-supervised pre-training approach, learning semantic features from contextually similar wafer bin maps (WBMs) before fine-tuning the model with labeled data for defect pattern prediction. In contrast, our proposed WaFixMatch model uses a saliency map derived from a CNN trained on labeled data to guide the semi-supervised learning process. This involves using weakly augmented labeled data and strongly augmented unlabeled data, with the saliency map informing the cutout augmentation strategy. This approach aims to enhance the model's ability to learn from unlabeled WBMs and improve defect pattern classification accuracy.

G. RESULTS AND DISCUSSION

We compared the proposed WaFixMatch with the WaPIRL [10] and Manivannan's methods [9], self-supervised and semi-supervised learning methods designed specifically for WBM defect classification. For the experiments, we adhered to the evaluation protocol presented by [10]. The main objective was to demonstrate the practical inference capability of WaFixMatch by observing the decrease in performance relative to the proportion of labelled data used. The results of WaPIRL [10] and supervised learning methods were consistent with the findings reported in their studies. Notably, the WaPIRL [10] method achieved the best among the augmentation techniques — Crop, Cutout, Noise, Rotation, and Shift—as detailed in Table 2 of the original paper. Manivannan [9] was implemented as described in Section IV-C, and we conducted experiments using the same number of encoders specified in the original paper. It is worth noting that in Table 3, standard deviations of WaPIRL [10] and Manivannan [9]'s methods were not presented because they just reported the optimal score, selected for each scenario.

Table 3 presents the results of the defect classification experiment on the WBM dataset, with the highest performance highlighted in bold. The proposed WaFixMatch showed an approximate improvement of 0.75 in the F1 score compared to Manivannan's method [9], and an improvement of about 0.49 compared to WaPIRL [10] in the most challenging setting where only 5% labeled dataset are used. Additionally, as the number of labeled images decreased, the proposed method exhibited a smaller decrease in the F1 macro score compared to other methods. This indicates that the proposed method demonstrates minimal performance degradation even when data is scarce, addressing a common issue in real-world applications. In contrast, supervised learning methods showed a steep decline in performance under label-scarce conditions, highlighting their dependency on a large amount of labeled data. These results are consistent with the findings of Sohn et al. [3], who demonstrated that consistency regularization and pseudo-labeling allow semi-supervised methods to use unlabeled data effectively, thereby reducing their dependency on labeled samples and maintaining high performance even in label-scarce scenarios. However, supervised learning methods, while achieving competitive performance when sufficient labeled data is available, face significant limitations when labels are sparse. Additionally, Our results align with Kong and Ni [8], demonstrating that semi-supervised methods enhance the decision boundary by leveraging unlabeled data, especially in wafer bin map datasets with imbalanced and limited labeled data. Although the proposed method ranked second in accuracy when 100% of labeled data was used, the score differences from the highest-performing method are not significant. In scenarios with full label usage, WaFixMatch showed a marginally lower F1-score than WaPIRL [10] (by 0.005). It is important to note that WaPIRL's [10] score was achieved under its best-performing pretext among five pretexts evaluated. This difference can be attributed to several factors. Firstly, WaPIRL's [10] self-supervised pre-training allows it to use robust feature representations, which can generalize well even when all labeled data is used, especially given the dataset's class imbalance. On the other hand, the specialized saliency-guided augmentations in WaFixMatch are primarily advantageous in low-data regimes, where preserving key features is critical. As the amount of labeled data increases, the relative impact of these augmentations may diminish. Finally, minor differences in hyperparameters, random initialization, and optimization can lead to slight variations in F1-score performance across the two methods.

TABLE 5. The classification performance reported using the macro F1 score and standard deviation (in parentheses) on WBM-811K under three different experiments with DenseNet-121 as the backbone model. The total count of labeled images used at each percentage, provided in square brackets. The best results are in bold.

Method	Proportion of Labeled Data (Number of Labeled Data Used)				
	5% [6,918]	10% [13,836]	25% [34,590]	50% [69,180]	100% [138,630]
Supervised	0.714	0.786	0.812	0.848	0.849
Manivannan's [9]	0.734	0.806	0.846	0.863	0.858
WaFixMatch (Proposed)	**0.793 (0.023)**	**0.817 (0.006)**	**0.851 (0.004)**	**0.872 (0.005)**	**0.868 (0.016)**

Table 4 demonstrates the effectiveness of each component in the proposed model to the overall performance. Four variants were compared: WaFixMatch (without KeepCutout and Rotation), WaFixMatch (without Rotation), WaFixMatch (without KeepCutout), and WaFixMatch (proposed). KeepCutout recognizes important areas when cutout using a saliency map and Rotation exploits rotational invariance only in weak augmentation. We observed the amount of change in the F1 score when two components were applied or not. As seen in Table 4, the two components are essential to increase the F1 score in every scenario. A significant difference in performance is evident when comparing scenarios with and without the application of KeepCutout, which is the main contribution of the proposed method. This demonstrates that applying cutout indiscriminately without regard for critical regions cause adverse leading to the loss of informative features essential for the identification of defective patterns.

TABLE 6. The classification performance reported using the macro F1 score and standard deviation (in parentheses) on WBM-811K under three different experiments with ResNet-50 as the backbone model. The total count of labeled images used at each percentage, provided in square brackets. The best results are in bold.

	Proportion of Labeled Data (Number of Labeled Data Used)				
Method	5% [6,918]	10% [13,836]	25% [34,590]	50% [69,180]	100% [138,630]
Supervised	0.629	0.676	0.785	0.847	0.870
WaPIRL [10]	0.794	0.823	0.857	0.874	**0.892**
WaFixMatch (Proposed)	**0.812 (0.022)**	**0.840 (0.017)**	**0.868 (0.001)**	**0.884 (0.005)**	0.884 (0.003)

Additionally, although Rotation in weak augmentation is not a primary proposal, it was included in the ablation study to highlight its distinct performance improvement compared to the general FixMatch approach. The comparative analysis of scenarios using Rotation augmentation versus those without it reveals performance gains, reinforcing that leveraging the rotational invariance characteristic inherent to WBM is beneficial. Applying both approaches helps achieve the best performance, indicating that exploiting rotational invariance and preserving important regions when applying cutout is crucial for WBM classification.

Lastly, Table 5 compares the proposed WaFixMatch with the supervised model and Manivannan's methods [9] using DenseNet-121 as the backbone model, while Table 6 compares WaFixMatch with the supervised model and WaPIRL [10] using ResNet-50 as the backbone model. In both tables, WaFixMatch consistently outperforms the baseline methods, achieving higher macro F1 scores across different labeled data proportions. For instance, with 5% labeled data, WaFixMatch achieves a macro F1 score of 0.793 with DenseNet-121 (Table 5) and 0.811 with ResNet-50 (Table 6), surpassing both the supervised model and baseline methods. These results indicate that WaFixMatch is particularly effective in low-data regimes using its unique augmentation and pseudo-labeling strategy to make efficient use of unlabeled data. While WaFixMatch performs robustly across most settings, we observe a slight difference in favor of WaPIRL [10] with 100% labeled data (by 0.008), consistent with earlier findings where WaPIRL's [10] self-supervised pre-training achieved marginally stronger results in fully labeled scenarios. This outcome aligns with WaPIRL's [10] strength in leveraging comprehensive labeled data, while WaFixMatch's wafer-specific augmentations continue to demonstrate significant advantages in low-data contexts. Overall, the consistently high performance of WaFixMatch, especially in scenarios with limited labeled data, highlights its robustness and effectiveness as a semi-supervised learning approach for wafer bin map classification.

V. CONCLUSION

In this study, we introduce an SSL framework specifically designed for WBM defect classification, incorporating a unique augmentation method. The proposed WaFixMatch uses a saliency map generated from a pre-trained model, enabling information-preserving augmentation of images during the SSL phase. This approach addresses the challenge of information loss in strongly augmented images by preserving critical regions. The experimental results demonstrate the effectiveness and practicality of the proposed method, particularly in managing scenarios with sparsely labeled data. The present study focuses on the challenge of applying strong augmentation without compromising crucial information, a limitation observed in our baseline models. The findings reveal significant performance improvements with our method, particularly in settings with limited labeled data. Furthermore, we examine the contribution of each component within WaFixMatch. The results shows that rotation augmentation preserves the inherent geometric properties of WBM images in weak augmentation, while strong augmentation using the saliency map preserves import regions pivotal for accurate defect classification. With continued exploration and enhancement, the proposed method shows the significant potential for practical application advancements. Future research focuses on developing augmentations to better understand the characteristics of WBMs and integrating advanced SSL techniques, presenting a promising tool for further elevating the performance of our framework.

REFERENCES

[1] K. Kyeong and H. Kim, "Classification of mixed-type defect patterns in wafer bin maps using convolutional neural networks," *IEEE Trans. Semicond. Manuf.*, vol. 31, no. 3, pp. 395–402, Aug. 2018, doi: 10.1109/TSM.2018.2841416.

[2] J. Kim, Y. Lee, and H. Kim, "Detection and clustering of mixed-type defect patterns in wafer bin maps," *IISE Trans.*, vol. 50, no. 2, pp. 99–111, Feb. 2018, doi: 10.1080/24725854.2017.1386337.

[3] K. Sohn, D. Berthelot, C. Li, Z. Zhang, N. Carlini, E. D. Cubuk, A. Kurakin, H. Zhang, and C. Raffel, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020.

[4] W. Lai, J. Huang, and M. Yang, "Semi-supervised learning for optical flow with generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 354–364.

[5] B. Li, F. Cheng, H. Cai, X. Zhang, and W. Cai, "A semi-supervised approach to fault detection and diagnosis for building HVAC systems based on the modified generative adversarial network," *Energy Buildings*, vol. 246, Sep. 2021, Art. no. 111044, doi: 10.1016/j.enbuild.2021.111044.

[6] D. Wang, Y. Zhang, K. Zhang, and L. Wang, "FocalMix: Semi-supervised learning for 3D medical image detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3950–3959, doi: 10.1109/CVPR42600.2020.00401.

[7] K. Clark, M.-T. Luong, C. D. Manning, and Q. Le, "Semi-supervised sequence modeling with cross-view training," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 1914–1925, doi: 10.18653/v1/d18-1217.

[8] Y. Kong and D. Ni, "Semi-supervised classification of wafer map based on ladder network," in *Proc. 14th IEEE Int. Conf. Solid-State Integr. Circuit Technol. (ICSICT)*, Oct. 2018, pp. 1–4, doi: 10.1109/ICSICT.2018.8564982.

[9] S. Manivannan, "An ensemble-based deep semi-supervised learning for the classification of wafer bin maps defect patterns," *Comput. Ind. Eng.*, vol. 172, Oct. 2022, Art. no. 108614, doi: 10.1016/j.cie.2022.108614.
[10] H. Kahng and S. B. Kim, "Self-supervised representation learning for wafer bin map defect pattern classification," *IEEE Trans. Semicond. Manuf.*, vol. 34, no. 1, pp. 74–86, Feb. 2021, doi: 10.1109/TSM.2020.3038165.
[11] T.-S. Li and C.-L. Huang, "Defect spatial pattern recognition using a hybrid SOM–SVM approach in semiconductor manufacturing," *Exp. Syst. Appl.*, vol. 36, no. 1, pp. 374–385, Jan. 2009, doi: 10.1016/j.eswa.2007.09.023.
[12] B. Kim, Y.-S. Jeong, S. H. Tong, I.-K. Chang, and M.-K. Jeongyoung, "A regularized singular value decomposition-based approach for failure pattern classification on fail bit map in a DRAM wafer," *IEEE Trans. Semicond. Manuf.*, vol. 28, no. 1, pp. 41–49, Feb. 2015, doi: 10.1109/TSM.2014.2388192.
[13] M. Saqlain, B. Jargalsaikhan, and J. Y. Lee, "A voting ensemble classifier for wafer map defect patterns identification in semiconductor manufacturing," *IEEE Trans. Semicond. Manuf.*, vol. 32, no. 2, pp. 171–182, May 2019, doi: 10.1109/TSM.2019.2904306.
[14] R. Wang and N. Chen, "Wafer map defect pattern recognition using rotation-invariant features," *IEEE Trans. Semicond. Manuf.*, vol. 32, no. 4, pp. 596–604, Nov. 2019, doi: 10.1109/TSM.2019.2944181.
[15] H. Lee, Y. Kim, and C. O. Kim, "A deep learning model for robust wafer fault monitoring with sensor measurement noise," *IEEE Trans. Semicond. Manuf.*, vol. 30, no. 1, pp. 23–31, Feb. 2017, doi: 10.1109/TSM.2016.2628865.
[16] T. Nakazawa and D. V. Kulkarni, "Anomaly detection and segmentation for wafer defect patterns using deep convolutional encoder–decoder neural network architectures in semiconductor manufacturing," *IEEE Trans. Semicond. Manuf.*, vol. 32, no. 2, pp. 250–256, May 2019, doi: 10.1109/TSM.2019.2897690.
[17] Y. Kong and D. Ni, "A semi-supervised and incremental modeling framework for wafer map classification," *IEEE Trans. Semicond. Manuf.*, vol. 33, no. 1, pp. 62–71, Feb. 2020, doi: 10.1109/TSM.2020.2964581.
[18] J. Yoo and S. Kang, "Class-adaptive data augmentation for image classification," *IEEE Access*, vol. 11, pp. 26393–26402, 2023, doi: 10.1109/ACCESS.2023.3258179.
[19] S. Kang, "Rotation-invariant wafer map pattern classification with convolutional neural networks," *IEEE Access*, vol. 8, pp. 170650–170658, 2020, doi: 10.1109/ACCESS.2020.3024603.
[20] I. Jeong, S. Y. Lee, K. Park, I. Kim, H. Huh, and S. Lee, "Wafer map failure pattern classification using geometric transformation-invariant convolutional neural network," *Sci. Rep.*, vol. 13, no. 1, p. 8127, May 2023, doi: 10.1038/s41598-023-34147-2.
[21] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," Tech. Rep., 2009.
[22] Y. Netzer, T. Wang, A. Coates, A. Bissacco, B. Wu, and A. Y. Ng, "Reading digits in natural images with unsupervised feature learning," in *Proc. NIPS*, 2011, pp. 1–4.
[23] A. Coates, A. Y. Ng, and H. Lee, "An analysis of single-layer networks in unsupervised feature learning," *J. Mach. Learn. Res.*, vol. 15, pp. 215–223, Jun. 2011.
[24] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[25] C. Gong, D. Wang, M. Li, V. Chandra, and Q. Liu, "KeepAugment: A simple information-preserving data augmentation approach," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1055–1064, doi: 10.1109/CVPR46437.2021.00111.
[26] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," 2013, *arXiv:1312.6034*.
[27] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "Randaugment: Practical automated data augmentation with a reduced search space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 3008–3017.
[28] D. Berthelot, N. Carlini, E. D. Cubuk, A. Kurakin, K. Sohn, H. Zhang, and C. Raffel, "ReMixMatch: Semi-supervised learning with distribution alignment and augmentation anchoring," in *Proc. 8th Int. Conf. Learn. Represent. ICLR*, Jan. 2019, pp. 1–13.
[29] M.-J. Wu, J. R. Jang, and J.-L. Chen, "Wafer map failure pattern recognition and similarity ranking for large-scale data sets," *IEEE Trans. Semicond. Manuf.*, vol. 28, no. 1, pp. 1–12, Feb. 2015, doi: 10.1109/TSM.2014.2364237.
[30] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. 7th Int. Conf. Learn. Represent.*, 2019.
[31] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, "Hyperband: A novel bandit-based approach to hyperparameter optimization," *J. Mach. Learn. Res.*, vol. 18, no. 1, pp. 6765–6816, Jan. 2017.

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